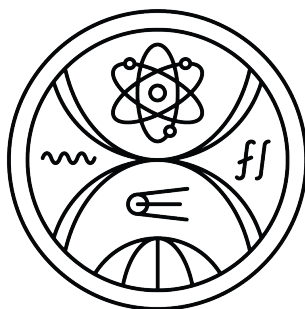


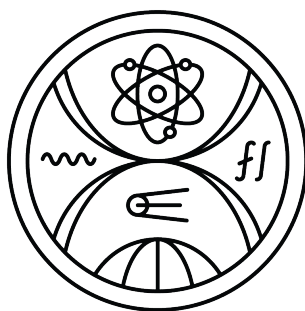
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



EXPLOITING ARTIFICIAL INTELLIGENCE IN SOFTWARE DEVELOPMENT

Master thesis

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



EXPLOITING ARTIFICIAL INTELLIGENCE IN SOFTWARE DEVELOPMENT

Master thesis

Study program: Applied informatics
Branch of study: Applied informatics
Department: Department of Applied Informatics
Supervisor: Ing. Lukáš Radoský



Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

Name and Surname: Bc. Kristián Rusnák
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Exploiting artificial intelligence in software development

Annotation: The advent of language models sparked great interest in their use in software development. Their ability to generate source code attracted particular attention. After the initial enthusiasm, attention is shifting more towards their use in other phases of the software development life cycle (SDLC), such as analysis, design, testing, and maintenance.

Choose one or more software development subtasks. Explore the possibilities of using artificial intelligence in these subtasks, especially the use of language models. Implement a solution for the selected subtask or subtasks based on artificial intelligence. Evaluate your solution on a suitable sample of data or users. Compare with existing solutions.

Aim: Design and implementation software development task solutions based on artificial intelligence.

Literature: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. <https://doi.org/10.1145/3695988>
[2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. <https://doi.org/10.1109/ICSE55347.2025.00112>
[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLOP '23). The Hillside Group, USA, Article 5, 1–31.

Keywords: artificial intelligence, large language models, software engineering, software development life cycle, SDLC

Supervisor: Ing. Lukáš Radoský
Department: FMFI.KAI - Department of Applied Informatics
Head of department: doc. RNDr. Tatiana Jajcayová, PhD.



Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

Assigned: 14.11.2025

Approved: 17.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
Guarantor of Study Programme

.....
Student

.....
Supervisor



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta: Bc. Kristián Rusnák
Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Exploiting artificial intelligence in software development
Využitie umelej inteligencie vo vývoji softvéru

Anotácia: Nástup jazykových modelov vyvolal veľký záujem o ich využitie vo vývoji softvéru. Veľkú pozornosť priťahovala najmä ich schopnosť generovať zdrojový kód. Po úvodnom nadšení sa pozornosť presúva viac k ich využitiu v iných fázach cyklu vývoja softvéru (SDLC), ako je analýza, návrh, testovanie, či údržba.

Zvoľte si jednu či viacero podúloh vývoja softvéru. Preskúmajte možnosti využitia umelej inteligencie pri týchto podúlohách, najmä však využitie jazykových modelov. Implementujte riešenie zvolenej podúlohy či podúloh využívajúce umelú inteligenciu. Svoje riešenie vyhodnoťte na vhodne zvolenej vzorke dát alebo používateľov. Porovnajte s existujúcimi riešeniami.

Cieľ: Návrh a implementácia riešení problémov vývoja softvéru pomocou umelej inteligencie.

Literatúra: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. <https://doi.org/10.1145/3695988>
[2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. <https://doi.org/10.1109/ICSE55347.2025.00112>
[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLOP '23). The Hillside Group, USA, Article 5, 1–31.

Kľúčové slová: umelá inteligencia, jazykové modely, softvérové inžinierstvo, cyklus vývoja softvéru

Vedúci: Ing. Lukáš Radoský
Katedra: FMFI.KAI - Katedra aplikovanej informatiky
Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

Spôsob sprístupnenia elektronickej verzie práce:
bez obmedzenia

Dátum zadania: 14.11.2025

Dátum schválenia: 17.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have written the entire master's thesis on the topic of 'Exploiting Artificial Intelligence in Software Development', including all its appendices and figures, independently, using the literature listed in the attached bibliography and artificial intelligence tools. I declare that I have used artificial intelligence tools in compliance with applicable laws, academic rights and freedoms, ethical and moral principles, while maintaining academic integrity, and that their use is appropriately acknowledged within the thesis.

Bratislava, 2026

.....
Bc. Kristián Rusnák

Acknowledgement

I would like to express my sincere gratitude to Ing. Lukáš Radoský for his continuous support, insightful advice, and expert guidance throughout the development of this thesis. His deep knowledge in the field of artificial intelligence and software development was instrumental in shaping this work. I would also like to thank my partner and my family, whose encouragement and patience throughout my academic journey meant more than words can express.

Abstract

Content of abstract.

Keywords: keywords 1, keywords 2, keywords 3

Abstrakt

Obsah abstraktu.

Klíčové slová: Klíčové slovo 1, Klíčové slovo 2, Klíčové slovo 3

Contents

1	Introduction	2
1.1	Large Language Models	2
1.2	Agentic AI	2
1.3	Log Analysis & Anomaly Detection	2
1.4	Key Principles (Generalizability, Robustness, Accessibility)	2
2	Related Work	3
2.1	Log Anomaly Detection	3
2.2	LLMs for Log Analysis	3
2.3	Agentic AI Systems	3
3	Dataset	4
3.1	Overview of Public Log Datasets (HDFS, BGL, Thunderbird)	4
3.2	Dataset Characteristics & Challenges	4
3.3	Data Augmentation	4
4	Implementation	5
4.1	Agentic Pipeline Design	5
4.2	Tool Design for the Agent	5
4.3	Model Selection & Configuration	5
5	Evaluation	6
5.1	Evaluation Methodology	6
5.2	Classification Metrics	6
5.3	Explanation Quality Metrics	6
5.4	Benchmark Results	6
6	Result Interpretation	7
6.1	Analysis of Classification Performance	7
6.2	Analysis of Explanation Quality	7
6.3	Limitations & Proposed Improvements	7

List of Figures

List of Tables

Terminology

Terms

- **Term 1**
Explanation 1.
- **Term 2**
Explanation 1.
- **Term 3**
Explanation 1.

Abbreviations

- **AAA** - Abbreviations 1.
- **BBB** - Abbreviations 2.
- **CCC** - Abbreviations 3.

Motivation

Motivation

Chapter 1

Introduction

1.1 Large Language Models

Large Language Models

1.2 Agentic AI

Agentic AI

1.3 Log Analysis & Anomaly Detection

Log Analysis & Anomaly Detection

1.4 Key Principles (Generalizability, Robustness, Accessibility)

Key Principles (Generalizability, Robustness, Accessibility)

Chapter 2

Related Work

2.1 Log Anomaly Detection

Log Anomaly Detection

2.2 LLMs for Log Analysis

LLMs for Log Analysis

2.3 Agentic AI Systems

Agentic AI Systems

Chapter 3

Dataset

3.1 Overview of Public Log Datasets (HDFS, BGL, Thunderbird)

Overview of Public Log Datasets (HDFS, BGL, Thunderbird)

3.2 Dataset Characteristics & Challenges

Dataset Characteristics & Challenges

3.3 Data Augmentation

Data Augmentation

Chapter 4

Implementation

4.1 Agentic Pipeline Design

Agentic Pipeline Design

4.2 Tool Design for the Agent

Tool Design for the Agent

4.3 Model Selection & Configuration

Model Selection & Configuration

Chapter 5

Evaluation

5.1 Evaluation Methodology

Evaluation Methodology

5.2 Classification Metrics

Classification Metrics

5.3 Explanation Quality Metrics

Explanation Quality Metrics

5.4 Benchmark Results

Benchmark Results

Chapter 6

Result Interpretation

6.1 Analysis of Classification Performance

Analysis of Classification Performance

6.2 Analysis of Explanation Quality

Analysis of Explanation Quality

6.3 Limitations & Proposed Improvements

Limitations & Proposed Improvements

Conclusion

Conclusion

Bibliography

- [1] Wei Guan, Jian Cao, Shiyu Qian, Jianqi Gao, and Chun Ouyang. LogLLM: Log-based anomaly detection using large language models. *arXiv preprint arXiv:2411.08561*, 2024. Submitted Nov 2024, last revised Apr 2025.
- [2] Juyoung Lee, Yeonsu Jeong, Taehyun Han, and Taejin Lee. LogRESP-Agent: A recursive AI framework for context-aware log anomaly detection and TTP analysis. *Applied Sciences*, 15(13):7237, 2025. Available at <https://doi.org/10.3390/app15137237>.
- [3] Jiaxing Qi, Shaohan Huang, Zhongzhi Luan, Shu Yang, Carol Fung, Hailong Yang, Depei Qian, Jing Shang, Zhiwen Xiao, and Zhihui Wu. Loggpt: Exploring chatgpt for log-based anomaly detection. 2024. Available at <https://ieeexplore.ieee.org/document/10466820>.
- [4] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. A prompt pattern catalog to enhance prompt engineering with ChatGPT. 2023.
- [5] Zeng Zhang, Wenjie Yin, and Xiaoqi Li. A novel GPT-based framework for anomaly detection in system logs. *arXiv preprint arXiv:2510.16044*, 2025. Submitted Oct 2025.
- [6] Zhiwei Zhang, Saifei Li, Lijie Zhang, Jianbin Ye, Chunduo Hu, and Lianshan Yan. LLM-LADE: Large language model-based log anomaly detection with explanation. *Knowledge-Based Systems*, 2025. Available at <https://doi.org/10.1016/j.knsys.2025.114064>.